

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

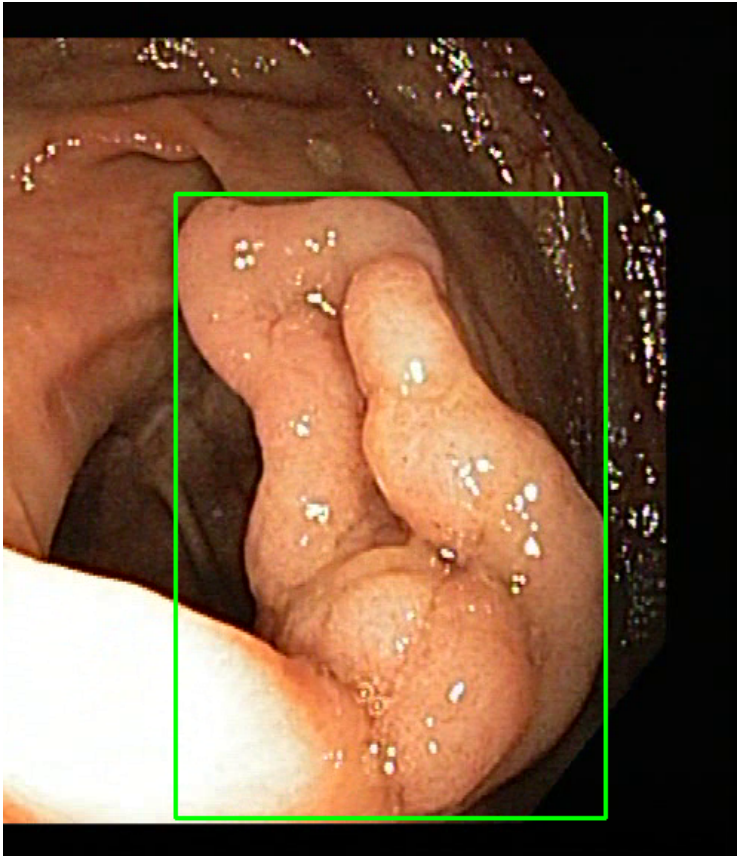
Video ID:	b4c522db-6034-4b94-bcdd-15fa74e86504
Total Frames:	1071
Frame Rate:	25.0 fps (assumed)
Duration:	42.8 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	426	✓
RT-DETR Detections	617	✓
MedSAM2 Detections	632	✓
Consensus (3-Model Agreement)	2	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	1	100.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	1	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 84 frames)



Feature	Value	Interpretation
Frame Index	1006	Frame 1006 of 1071
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 Unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 83.0%
Redness Score	0.174	Color-based vascularization
Relative Radius	38.4%	Polyp size as % of frame diagonal
Texture Score	0.138	Surface morphology
Vessel Visibility	0.104	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	87.2%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 1 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 1 (100.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.